257 A Robust Method for Automatic and Dynamic Action Prediction of a User Using Inductive Learning with N-gram

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1. Introduction

Recently, baby birthrate is decreasing and it seems that the society is aging. Caring for the aged or bodily inconvenient people is under pertinent requirement. For the aged or bodily inconvenient persons, a caring system instead of manually care needs to be developed. Incase such a system is planned. The most important problem is how to predict a user's intention and action.

Various learning approaches will be planned to the action prediction system. The analytical approach needs to give the prepared rules adapted to the user and the adaptability of this method is low. The problem is that the custom and taste will change with each user. Consequently dynamic adaptation of a system cannot be realized.

We can also consider it from statistical approach, The following problems may come out. They need the huge database of user's action in order to acquire a good prediction result. Moreover, it is difficult to give a lot of examples beforehand for each user. It involves a lot of time and heavy labor preparing so many examples for each user. For the others, it is a fault that the exception examples cannot be coped with these methods. Accordingly, predicting the intention from a user's actual action's history by Inductive Learning is desirable.

Inductive Learning can acquire inherent rule through extracting a common part and a difference part recursively from examples [1],

Following merits appears using the Inductive Learning: First, the characteristic of Inductive Learning is that it can be learned with sufficient accuracy even from a small number of data. It is possible to acquire the rules including the exception example. Next, because it is learned from the data history while the system can be acquired automatically, it is unnecessary to prepare a huge database for each user beforehand. Moreover, because the data is different with each user, the extracted rules will be different so that the dynamic adaptation is possible to each user. In short, this method is robust. The system based on it can adapt to a user dynamically by acquiring rules from examples using the Inductive Learning [2].

Each rule acquired by Inductive Learning is estimated by the likelihood. In order to evaluate the whole relations of the rules, we introduce the N-gram used to our system. If we consider from an action sequence of a user, it is required to analyze relation of action in alignment with the passage of time for the system. On the other hand, it is the script, which expresses the standardized action that a person performs ordinary things in the form of the chain of a phenomenon in alignment with the passage of time. It is considered from the analysis of a script using N-gram that a user's action can be predicted. We propose this method is the action prediction of a user using Inductive Learning with N-gram [3].

2. Outlines

The system based on this method is a room with learning functions. For easy analysis, the component of the room consists of eight artificial elements and four natural elements. The changed coefficient of the state shows what a user does before. It is assumed that the system can acquire the state of the room automatically with actions of the user. Each element has some values given beforehand

2.1.Description of causal relationship

Natural element (N) is symbolized as follows:

$N_k^n = | W_k^n | H_k^n | M_k^n | B_k^n | T_k^n |$ *1

Similarly, artificial element (ART) is symbolized as follows:

$ART_k^n = | D_k^n | TV_k^n | P_k^n | C_k^n | L_k^n | AC_k^n | TEL_k^n | BELL_k^n |$

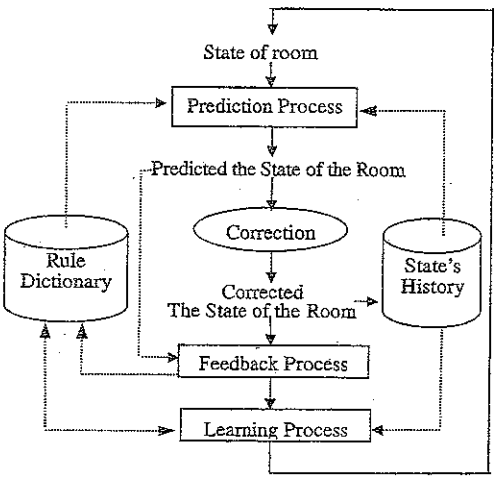
The result that changed the state of the artificial element can be treated as a user's reaction (REA_k^n). The cause(Ca) used as action of a user is defined as follows:

$Ca_k^n = N_k^n \,||\, ART_k^n$ *2

The result (REA_k^n) of changed state of a user's action and a corresponding artificial element reacts to an intention of a user. Using this result, we can find the causal relation such as the $Ca_k^n \rightarrow REA_k^n$ exists. We define $Ca_k^n \rightarrow REA_k^n$ as a rule with causal relation.

2.2. OUTLINE OF SYSYTEM

Figure 1: Procedure



*1 "n" expresses the order of the data registered into the state history. "k" expresses the date. "|" is a symbol to distribute about the state of each element. The natural elements (N) consist of hour (H), minute (M), brightness (B), day of week (W), and temperature (T). The eight artificial elements consist of a door (D), TV, TV program (P), curtain(C), light (L), an air-conditioner (AC), a telephone (TEL), and a bell (B) in this system.

*2: "| |" is a symbol to distribute about a classification of each symbolized element.

Figure 1 shows an outline of the system based on this method. The procedure consists of the prediction process, the correction process, the feedback process and the learning process in this order.

The rules in the rule dictionary are acquired in the learning process, and it is classified into four ranks that are M (CR≥90%), H (70%≤CR< 90%), C (30%≤CR< 70%) and L (CR< 30%). The rules are applied in this order.

3. EVALUATION EXPERIMENT

3.1. EVALUATION OF RULES

The rule's credibility evaluation function CEF is defined as:

$$CEF = \alpha P(A_{n+1}|A_n) + \beta CR - \gamma ER \quad \cdots (1)$$

, Where α, β and γ are coefficients. CR is the rate of the correct prediction. ER is the rate of the erroneous prediction. $P(A_{n+1}|A_n)$ is the probability of the neighboring action. CR and ER are based on the rule dictionary and they are updated in the feedback process. $P(A_{n+1}|A_n)$ is based on the state's history. The credibility for the rule is higher when $P(A_{n+1}|A_n)$ is higher, and CR is higher and ER is lower. The CR and ER defined as (2) and (3):

$$CR = \frac{CF}{CF + EF} \quad \cdots (2) \qquad ER = \frac{EF}{CF + EF} \quad \cdots (3)$$

,where CF is the appearance degree for the correct prediction, EF is the appearance degree for the erroneous prediction. In the feedback process, the predicted result is compared with the corrected result. According this result, the credibility of the rules, CF, EF and CR. And rules rank is updated. The certainty degree for the rule in the rule dictionary is updated, too. It is judged by comparing the result of prediction and its corrected result. In the learning process, the extraction of recursive processing of the initial rule is performed. It is necessary to give the suitable degree of abstraction. We define the definition of the rule abstraction degree such as:

$$DA=\left(\frac{nature_element_variables_number}{nature_element_number} + \frac{artificial_element_variables_number}{artificial_element_number}\right)\times 100\% \quad \cdots (4)$$

,where DA is the abstraction degree of the rule. At the same time, from the data history, the script of action that the user performed every day is generated. From the registered script, neighboring action probability $P(A_{n+1}|A_n)$ is analyzed by Bi-gram and the analyzed result is registered into the rule dictionary. Thus, the system based on this method is improved by the repetition of these processes.

3.2 DATA AND PROCEDURE

The system based on this method has been developed for an experiment to evaluating the adaptability of this method for a user

The data used in this experiment is based on the actual life of the first author. The number of inputted states is 945. The experiment period was nine weeks. In learning process, we consider a range of the natural elements as the judgment standard of the different part and the common part. In the formula (1) the parameters α, β, γ were obtained in the preliminary experiment that is 100, 1 and 10 respectively. The rule dictionary and the state's history are empty at first.

3.3 RESULTS

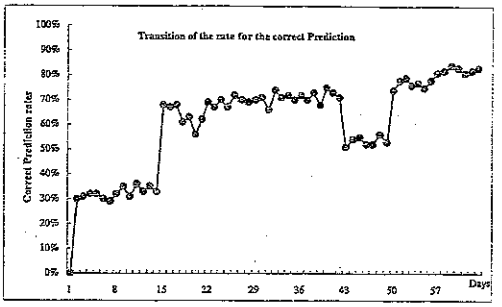


Figure 2 the rate of the Correct Prediction

Figure 2 shows the results in the rate of the correct prediction. In Figure 2, the rate of the correct prediction is increasing as the number of the inputted room's states is increasing. The correct rate is about 83[%] in the final. The rate of week's average prediction was 60.1[%].

It was shown that the system in this method carries out dynamic adaptation to a user. That is to say, the system can be learned according to custom and taste of the user. Although the right prediction rate fell once according to a user's new life but the rules are also gained. It is clear that the correct prediction rate increased gradually.

4. CONCLUSIONS

This method acquires rules from the state history automatically by its learning capability. It is not necessary to prepare the rule united or a lot of examples for the user beforehand. It is learned from the example that the system was gained. The rule is acquired according to custom and taste of a user by its own adaptability. The result of the experiment shows that this method is effective for the user.

Future works consist of the following contents: the check of the flexibility of the system using two or more users, automatic adaptation of the range in the natural element's state and use of the Maximum Entropy Model.

5. REFERENCES

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